

Wilson Chow

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Professional Summary

Software developer with 5 years of experience, centered on frontend (React, TypeScript, Figma) with strong backend and cloud depth. Ship end-to-end products: design systems, headless CMS/search, tRPC APIs, and SST/AWS infra. Certified AWS Solutions Architect Associate. Building production AI agents and RAGs with Mastra, AI SDK, and Langchain.

Experience

Software Developer, Vioo Company Limited, May 2022 - Present

- Developed and maintained 15+ production React / React Native apps end-to-end, including Figma UI/UX and design systems.
- Partnered with clients and stakeholders to scope work, iterate on UX, and deliver production releases.
- Established a shared component/hook library (Radix, Tailwind CSS) with Storybook / Fumadocs / MDX docs used across multiple client apps.
- Created advanced Framer Motion animations and micro-interactions (scroll-based, animate presence, layout transitions) across client apps.
- Built production AI agents (workflows, memory, workspaces, observability) with Mastra, AI SDK, and Langchain; shipped reusable agent chat and tool-invocation UI.
- Replaced legacy WordPress with headless CMS (Strapi / Sanity) and search (Meilisearch / Algolia). Cut load time by 80%, search latency by 90%, and hit PageSpeed Insights >90.
- Built typed APIs with Hono, Drizzle, OpenAPI, and tRPC on AWS Lambda / API Gateway, backed by Postgres and DynamoDB, with Scalar API docs.
- Adopted SST for app deployment and infra, cutting build and deploy time by 50%+.
- Led frontend code and performance reviews — composable React patterns, route-level code splitting, lazy-loaded heavy UI — and added Cypress E2E coverage for critical flows.

Selected Projects

Multi-channel Durable Agent, Apr 2026 - Present

- Led a multi-channel durable agent platform (web + Discord), clients can drop and reconnect mid-stream.
- Built workflow, memory, scheduling, and workspace primitives — pause/resume, retries, file storage, sandboxed tools, and skills — on Mastra and AI SDK.
- Shipped RAG end-to-end: ingest/parse, chunking, Pinecone retrieval with query/rerank (and graph query where needed).
- Hono + Better Auth backend on ECS; Postgres for core data, S3 as filesystems.
- Built agent observability and evals: run/tool traces and cost/latency metrics in ClickHouse, plus scored eval suites (golden prompts + LLM-as-judge) for quality regressions.
- Designed and shipped a custom @mention file tagging component for agent prompt input using TipTap.

Incremental Static Regeneration Next.js Site, Nov 2025

- Built a Next.js content site with Strapi backend and Meilisearch, hitting LCP < 0.5s, CLS < 0.1, and PageSpeed Insights >90.
- Hybrid ISR: main pages and the latest 100 articles pre-rendered at build; remaining pages generated on demand as fallbacks.
- Tag-based revalidation via Strapi content webhooks so publishes update live pages without full rebuilds.
- Interactive UI with TanStack Query server prefetch/hydration; forms via TanStack Form, Zod, and Query mutations.
- Turborepo deployed to AWS ECS with SST, routed through API Gateway; media on S3.
- Accessibility: semantic landmarks/headings, keyboard-accessible nav and forms, visible focus, and meaningful alt text for CMS media.
- SEO: metadata/OG tags from Strapi, canonical URLs, sitemap + robots.txt, and structured data (Article JSON-LD) for indexable content pages.

React Native App, April 2024

- Led Figma design and React Native frontend for a catering-brand membership app (digital card, points, offers).
- Integrated typed REST clients from OpenAPI/Swagger with backend; shipped iOS & Android to App Store and Play Store.
- Push + deep-linked campaigns for promos and milestones.
- Maintained >99.5% crash-free sessions post-launch, monitored with Sentry.
- Cut list jank on menus/history with FlashList (recycling, tuned batch/window) instead of ScrollView-mapped rows.
- Kept inactive tabs cheap with react-native-screens freeze + deferred screen mounts; reserved Reanimated for native-thread transitions.
- Cached remote images in feeds and avoided full-resolution assets in long lists.

Other Projects

I also developed a range of applications:

- Health & wellness RAG agent (Next.js, Langchain, Shopify product retrieval)
- Restaurant loyalty app (React Native)
- Membership portal for a business organization (React Native)
- Event registration site (Next.js)
- Campaign / marketing sites (Next.js, Sanity / Strapi)
- Resource management system with RBAC (Next.js)
- E-commerce storefront (Next.js, Shopify GraphQL)
- Homebound elderly care app (Flutter)
- Book ordering system (Laravel, React)

Skills

- **Programming Languages:** Typescript, Python
- **Frontend:** React, Next.js, React Native, Expo, Tanstack Query, Framer Motion, Tailwindcss, Shadcn UI, Storybook, Figma, MDX
- **Backend:** Hono, Drizzle, OpenAPI, tRPC, Scalar, Pinecone, Meilisearch, Strapi, Sanity
- **AI:** Mastra, AI SDK, Langchain, Strands
- **Testing:** Cypress, Vitest, React Testing Library
- **Database:** Postgres, DynamoDB
- **DevOps:** Docker, SST, AWS, Vercel, Turborepo

Certifications

- AWS Solutions Architect Associate, 2026
- AWS Cloud Practitioner, 2025

Education

Univeristy College London

- MPharm Pharmacy, 2016 - 2020
- First class honours
- Registered pharmacist (GPhC), 2021 - present